Explanation of the indicators and thresholds chosen

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1 Explanation of the LCA indicators

Life Cycle Assessment (LCA) mobilises several environmental indicators, called impact categories, in order to cover all the pressures exerted on the environment. The ADEME report on GPUs for artificial intelligence [1] retains sixteen indicators, selected from the PEF (Product Environmental Footprint) methodology of the European Commission [2]. These indicators fall into two main families: problem-oriented impact indicators (*mid-point*), which reflect an environmental pressure halfway along the cause-effect chain, and flow indicators, which directly quantify resources consumed or waste produced. Each indicator is expressed in a specific unit and relies on a recognised assessment model. The sixteen indicators are described below.

1.1 Climate change (GWP)

Climate change is measured by the global warming potential over 100 years (GWP100), expressed in kilograms of carbon dioxide equivalent (kg CO_2 eq). It quantifies the contribution of greenhouse gases to the increase in the planet's average temperature. The model used is that of the IPCC [3]. This indicator is classified at recommendation level I, i.e. the most robust in the PEF methodology [2].

1.2 Ozone layer depletion (ODP)

Ozone layer depletion is measured by the ozone depletion potential (ODP), expressed in kilograms of CFC-11 equivalent (kg CFC-11 eq). It reflects the destruction of stratospheric ozone by halogenated substances. The model used is that of the World Meteorological Organization [4]. This indicator is classified at recommendation level I [2].

1.3 Fine particulate matter emissions (PM)

Fine particulate matter emissions are measured by their impact on human health, expressed as disease incidence. These are particles with a diameter of less than 10 microns, whose inhalation causes respiratory and cardiovascular disorders. The model used is the one recommended by UNEP [5]. This indicator is classified at recommendation level I [2].

1.4 Acidification (AP)

Acidification is measured by the accumulation of proton equivalents (Accumulated Exceedance), expressed in moles of hydrogen ion equivalent (mol H^+ eq). It reflects the deposition of acidifying compounds (SO_2, NO_x, NH_3) in soils and waters. The model used is that of Seppälä et al. [6] and Posch et al. [7]. This indicator is classified at recommendation level II [2].

1.5 Freshwater aquatic eutrophication (Epf)

Freshwater eutrophication is measured by the fraction of nutrients reaching the final compartment, expressed in kilograms of phosphorus equivalent (kg P eq). It reflects the excessive enrichment of aquatic environments with phosphorus, responsible for algal blooms and oxygen depletion. The model used is the EUTREND model [8], as implemented in ReCiPe. This indicator is classified at recommendation level II [2].

1.6 Marine aquatic eutrophication (Epm)

Marine eutrophication is measured by the fraction of nutrients reaching the marine compartment, expressed in kilograms of nitrogen equivalent (kg N eq). It reflects the excessive enrichment of marine environments with nitrogen. The model used is also the EUTREND model [8]. This indicator is classified at recommendation level II [2].

1.7 Terrestrial eutrophication (Ept)

Terrestrial eutrophication is measured by the accumulation of nitrogen equivalents (Accumulated Exceedance), expressed in moles of nitrogen equivalent (mol N eq). It reflects the excessive enrichment of soils with nitrogen, which alters the composition of terrestrial ecosystems. The model used is that of Seppälä et al. [6] and Posch et al. [7]. This indicator is classified at recommendation level II [2].

1.8 Ionising radiation, human health (IR)

Ionising radiation is measured by the relative human exposure efficiency relative to Uranium 235, expressed in kilobecquerels of U-235 equivalent (kBq U-235 eq). It reflects human exposure to radionuclides released during various activities, notably nuclear power generation. The model used is that developed by Dreicer et al. [9] and Frischknecht et al. [10]. This indicator is classified at recommendation level II [2].

1.9 Photochemical ozone formation, human health (POCP)

Photochemical ozone formation is measured by the increase in tropospheric ozone concentration, expressed in kilograms of non-methane volatile organic compounds equivalent (kg NMVOC eq). It reflects the formation of ground-level ozone, harmful to human health and ecosystems. The model used is the LOTOS-EUROS model [11], as implemented in ReCiPe. This indicator is classified at recommendation level II [2].

1.10 Human toxicity, carcinogenic effects (CTUh-c)

Carcinogenic human toxicity is measured by the Comparative Toxic Unit for humans, expressed in CTUh. It reflects the potential increase in cancer cases linked to exposure to toxic chemical substances. The model used is USEtox [12]. This indicator is classified at recommendation level III/interim [2], which means that its robustness is lower than that of the previous indicators.

1.11 Human toxicity, non-carcinogenic effects (CTUh-nc)

Non-carcinogenic human toxicity is measured by the same unit as the previous indicator (CTUh). It reflects health effects other than cancer (neurological, hepatic, renal disorders, etc.) linked to exposure to chemical substances. The model used is also USEtox [12]. This indicator is classified at recommendation level III/interim [2].

1.12 Ecotoxicity, freshwater (CTUe)

Freshwater ecotoxicity is measured by the Comparative Toxic Unit for ecosystems, expressed in CTUe. It reflects the potential impact of chemical substances on organisms living in freshwater aquatic environments. The model used is USEtox [12]. This indicator is classified at recommendation level III/interim [2].

1.13 Water use (WU)

Water use is measured by the user deprivation potential, expressed in cubic metres of world water equivalent (m^3 world eq). This indicator does not measure an actual volume of water, but a volume weighted by the scarcity of water in the region where it is withdrawn. The model used is AWARE 100 [13], recommended by UNEP [5]. This indicator is classified at recommendation level III [2].

1.14 Abiotic resource depletion, minerals and metals (ADPe)

Mineral and metal resource depletion is measured by the abiotic depletion potential of ultimate reserves, expressed in kilograms of antimony equivalent (kg Sb eq). It reflects the decrease in available reserves of mineral and metal resources. The model used is CML 2002 [14, 15]. This indicator is classified at recommendation level III [2].

1.15 Abiotic resource depletion, fossil fuels (ADPf)

Fossil resource depletion is measured by the abiotic depletion potential of fossil fuels, expressed in megajoules (MJ). It reflects the consumption of primary energy from non-renewable sources (oil, natural gas, coal, uranium). The model used is CML 2002 [14, 15]. This indicator is classified at recommendation level III [2].

1.16 Land occupation / Soil quality (LU)

Land occupation is measured by a soil quality index, expressed in points (pt). This index aggregates four sub-indicators: biotic production, erosion resistance, mechanical filtration and groundwater recharge. The model used is the soil quality index based on LANCA [16, 17]. This indicator is classified at recommendation level III [2]. Note that the ADEME report [1] judged this indicator to be of little relevance for digital technology and did not retain it in its results.

2 Thresholds for each indicator per month per inhabitant

2.1 Planetary boundaries

The thresholds chosen were selected in accordance with the ADEME LCA report on GPUs for artificial intelligence [1]. They reuse the objectives of the Stockholm Resilience Centre [18], which identifies 9 planetary boundaries for living in a sustainable ecosystem, i.e. one that avoids abrupt and unpredictable changes in the planetary environment.

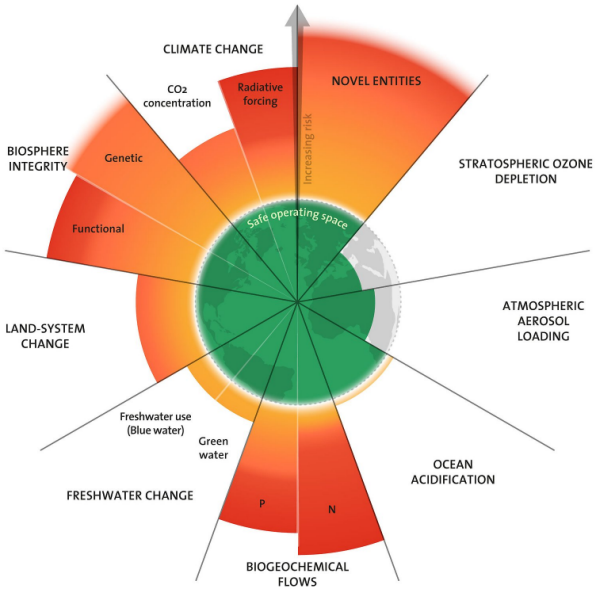


Figure 1: Planetary boundaries and their overshoot in 2025 [18]

2.2 Sustainable thresholds per LCA indicator

From the planetary boundaries, the Joint Research Centre [19] defined a threshold per LCA category not to be exceeded.

Catégorie d'impact	Facteur de limite planétaire	Unité
Changement climatique	6,81E+12	kg CO_2 eq./limite planétaire
Appauvrissement de la couche d'ozone	5,39E+08	kg CFC-11 eq./limite planétaire
Émissions de particules fines	5,16E+05	disease incidences/limite planétaire
Acidification	1,00E+12	mol H+ eq./limite planétaire
Eutrophisation aquatique, eaux douces	5,81E+09	kg P eq./limite planétaire
Eutrophisation aquatique, marine	2,01E+11	kg N eq./limite planétaire
Eutrophisation, terrestre	6,13E+12	mol N eq./limite planétaire
Rayonnements ionisants, santé humaine	5,27E+14	kBq U-235 eq./limite planétaire
Formation d'ozone photochimique, santé humaine	4,07E+11	kg NMVOC eq./limite planétaire
Toxicité humaine, effets non cancérigènes	4,10E+06	CTUh/limite planétaire
Épuisement des ressources abiotique, combustibles fossiles	2,24E+14	MJ/limite planétaire
Épuisement des ressources abiotique, minéraux et métaux	2,19E+08	kg Sb eq./limite planétaire
Utilisation de l'eau	1,82E+14	m3 water eq of deprived water/limite planétaire
Ecotoxicité, eaux douces	1,31E+14	CTUe/limite planétaire
Toxicité humaine, effets cancérigènes	9,62E+05	CTUh/limite planétaire

Figure 2: Weighting factor according to planetary boundaries (extract from ADEME [1])

In order to bring the results to a more intelligible scale, each of the planetary boundaries is expressed per inhabitant. The boundary must then be understood as a form of average annual budget per inhabitant of the world. Considering a population of 8.06E+09 inhabitants, the boundaries transcribed in the following table are obtained (extract from ADEME [1]).

Catégorie d'impact	Facteur de limite planétaire	Unité
Changement climatique	8,45E+02	kg CO_2 eq./limite planétaire/habitant
Appauvrissement de la couche d'ozone	6,69E-02	kg CFC-11 eq./limite planétaire/habitant
Émissions de particules fines	6,40E-05	disease incidences/limite planétaire/habitant
Acidification	1,24E+02	mol H+ eq./limite planétaire/habitant
Eutrophisation aquatique, eaux douces	7,21E-01	kg P eq./limite planétaire/habitant
Eutrophisation aquatique, marine	2,49E+01	kg N eq./limite planétaire/habitant
Eutrophisation, terrestre	7,60E+02	mol N eq./limite planétaire/habitant
Rayonnements ionisants, santé humaine	6,54E+04	kBq U-235 eq./limite planétaire/habitant
Formation d'ozone photochimique, santé humaine	5,05E+01	kg NMVOC eq./limite planétaire/habitant
Toxicité humaine, effets non cancérigènes	5,09E-04	CTUh/limite planétaire/habitant
Épuisement des ressources abiotique, combustibles fossiles	2,78E+04	MJ/limite planétaire/habitant
Épuisement des ressources abiotique, minéraux et métaux	2,72E-02	kg Sb eq./limite planétaire/habitant
Utilisation de l'eau	2,26E+04	m3 water eq of deprived water/limite planétaire/habitant
Ecotoxicité, eaux douces	1,62E+04	CTUe/limite planétaire/habitant
Toxicité humaine, effets cancérigènes	1,19E-04	CTUh/limite planétaire/habitant

Figure 3: Weighting factor according to planetary boundaries per year per inhabitant (extract from ADEME [1])

2.3 Actual consumption per LCA indicator

It is important to understand that these thresholds do not represent the actual consumption of each inhabitant. For some criteria, the sustainable thresholds are exceeded and actual consumption per inhabitant is too high; for others, consumption per inhabitant is below the thresholds per inhabitant. It is therefore possible to extract a table with the theoretical thresholds per year per inhabitant, the actual consumption per year per inhabitant, and whether or not the "annual budget per inhabitant" for each indicator is exceeded.

Indicateur	Seuil soutenable	Consommation réelle	Ratio	Statut
Particules fines (disease inc.)	6.40e-05	6.37e-04	9.95	DÉPASSÉ
Changement climatique (kg CO2 eq)	845.00	7.76e+03	9.18	DÉPASSÉ
Eutrophisation eau douce (kg P eq)	0.72	2.55	3.54	DÉPASSÉ
Ressources fossiles (MJ)	2.78e+04	6.53e+04	2.35	DÉPASSÉ
Ressources minérales (kg Sb eq)	0.03	0.06	2.13	DÉPASSÉ
Eutrophisation marine (kg N eq)	24.90	28.30	1.14	DÉPASSÉ
Toxicité humaine non-cancer (CTUh)	5.09e-04	4.75e-04	0.93	OK
Ozone photochimique (kg NMVOC eq)	50.50	40.60	0.80	OK
Écotoxicité eau douce (CTUe)	1.62e+04	1.18e+04	0.73	OK
Utilisation de l'eau (m3 world eq)	2.26e+04	1.15e+04	0.51	OK
Acidification (mol H+ eq)	124.00	55.50	0.45	OK
Appauvrissement ozone (kg CFC-11 eq)	0.07	0.02	0.35	OK
Toxicité humaine cancer (CTUh)	1.19e-04	3.85e-05	0.32	OK
Eutrophisation terrestre (mol N eq)	760.00	177.00	0.23	OK
Radiations ionisantes (kBq U235 eq)	6.54e+04	4.22e+03	0.06	OK

Figure 4: Sustainable thresholds per year per inhabitant (column 2) (JRC/ADEME [19, 1]) vs actual consumption per year per inhabitant (column 3) (PEF EU [2])

Obviously, the fact that the annual budget per inhabitant for an indicator is not exceeded does not mean that efforts should not be made to minimise it. This can easily be seen with carcinogenic human toxicity: even if the planetary threshold is far from being reached, an increase in this impact remains very negative. The reason is simple: planetary boundaries reflect thresholds for the stability of the Earth system, not acceptable levels of health impact. In other words, just because the budget is not exceeded does not mean that cancer causes no harm. It directly affects populations, unequally, and any increase must therefore be avoided.

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